

Heston Model: Simulation and Option Pricing Analysis

Author: Pandolfo Edoardo

Course: Stochastic Methods For Finance

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Software: Python (Implementation via Full Truncation Scheme)

1. Introduction

The **Heston model** [1](#) represents a cornerstone in quantitative finance, addressing the limitations of the Black-Scholes framework by introducing stochastic volatility. Unlike the constant volatility assumption, the Heston model describes the instantaneous variance as a mean-reverting process. This feature allows the model to capture empirical market phenomena such as **volatility clustering**, **fat-tailed return distributions**, and the **implied volatility smile** observed in option markets.

2. Model Dynamics and Parameters

2.1 Heston Dynamics

Under the risk-neutral measure $\mathbb{Q}$, the joint evolution of the asset price S_t and the instantaneous variance v_t is governed by the following coupled stochastic differential equations:

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{v_t}S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v\end{aligned}$$

where dW_t^S and dW_t^v are correlated Brownian motions with instantaneous correlation ρ , i.e., $dW_t^S dW_t^v = \rho dt$. The parameters have the following financial interpretation:

- κ (kappa): rate of mean reversion of the variance process towards its long-run mean.
- θ (theta): long-run mean of the variance (so that $\sqrt{\theta}$ is the long-run volatility level).
- σ (sigma): volatility of volatility, controlling the randomness of the variance process itself.
- ρ (rho): correlation between asset return and variance shocks. Negative values produce the leverage effect.
- v_0 : the initial variance at time $t = 0$.

2.2 Simulation Parameters

The parameters used throughout this report are summarized in the table below. Note that $\theta = 0.04$ corresponds to a long-run volatility of 20%.

Parameter	Symbol	Value	Description
Initial Asset Price	S_0	100	Current stock price
Risk-free Rate	r	0.02	Continuously compounded
Time Horizon	T	1.0 yr	Maturity
Mean Reversion Speed	κ	2.0	Heston parameter
Long-run Variance	θ	0.04	(20% vol)
Volatility of Volatility	σ	0.5	Heston parameter
Correlation	ρ	-0.7	Leverage effect
Initial Variance	v_0	0.04	Heston parameter
Time Steps	N	250	Daily steps
Simulation Paths	M	10,000	Monte Carlo paths

3. Task 6 – Feller Condition and Full Truncation

3.1 The Feller Condition

A key theoretical question for the Heston model concerns whether the variance process can reach zero. The **Feller condition** [2] provides a sufficient criterion for the variance to remain strictly positive almost surely:

$$2\kappa\theta > \sigma^2$$

With the base parameters of this homework, we have:

$$2 \times 2.0 \times 0.04 = 0.16 \quad \text{vs.} \quad \sigma^2 = 0.5^2 = 0.25$$

Since $0.16 < 0.25$, the Feller condition is **violated**. This means the CIR-type variance process can touch zero and, in a naive discretization, may produce negative values. This is not just a numerical artifact: it has real economic consequences, since a negative variance is undefined in the context of the diffusion term $\sqrt{v_t}$.

3.2 Full Truncation Scheme

To handle this issue in the Euler-Maruyama discretization, we adopt the **full truncation scheme** proposed by Lord et al. (2010) [3]. The discretized variance update becomes:

$$v_{t+1} = v_t + \kappa(\theta - v_t^+) \Delta t + \sigma \sqrt{v_t^+ \Delta t} Z^v, \quad v_t^+ = \max(v_t, 0)$$

This scheme applies the truncation $\max(v, 0)$ inside the drift term and inside the diffusion coefficient as well as in the asset price dynamics. The approach is conservative but numerically stable and widely used in practice.

Implementation note: Task 6 was implemented first in the notebook, and the full truncation was incorporated directly into the core simulation function `heston_model_sim()`, which is reused in all subsequent tasks. A separate diagnostic simulation confirmed that, across 10,000 paths with 250 time steps, approximately 24,887 attempted-negative variance steps occurred before truncation, involving a significant fraction of paths – consistent with the violated Feller condition.

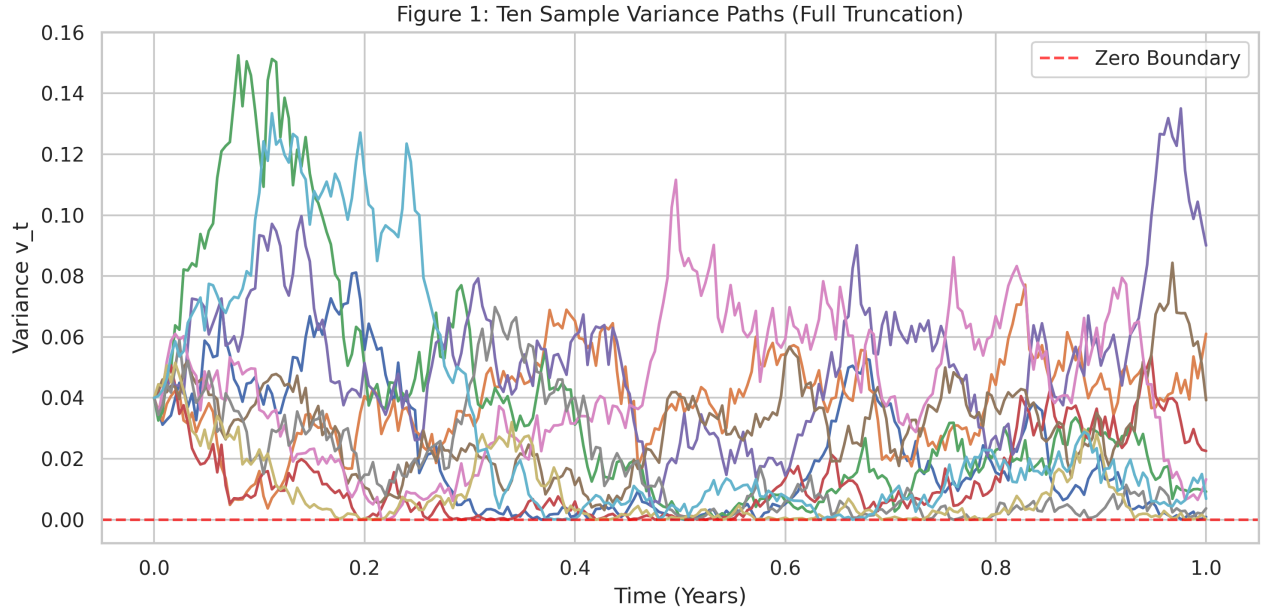


Figure 1: Ten sample variance paths under full truncation. The red dashed line marks zero. Paths are reflected upwards whenever they touch the boundary, illustrating the truncation mechanism.

4. Task 1 – Simulation of Asset Price and Variance Paths

Using the full truncation scheme described above, we simulate $M = 10,000$ joint paths of (S_t, v_t) over a one-year horizon with $N = 250$ daily time steps. The two Brownian motions are generated as correlated standard normals via a Cholesky decomposition of the correlation matrix with $\rho = -0.7$.

The asset price update follows a log-Euler scheme to preserve positivity of S_t :

$$S_{t+1} = S_t \times \exp \left[\left(r - \frac{1}{2} v_t^+ \right) \Delta t + \sqrt{v_t^+ \Delta t} Z^S \right]$$

Figure 2 shows ten representative sample paths of the asset price (left panel) and the variance process (right panel). The asset price paths display typical log-normal-like behavior with stochastic amplitude of fluctuations driven by the time-varying variance. The variance paths exhibit mean-reversion towards $\theta = 0.04$ (20% long-run volatility), with occasional excursions that are truncated at zero.

Figure 2: Joint Simulation of Asset Price and Variance

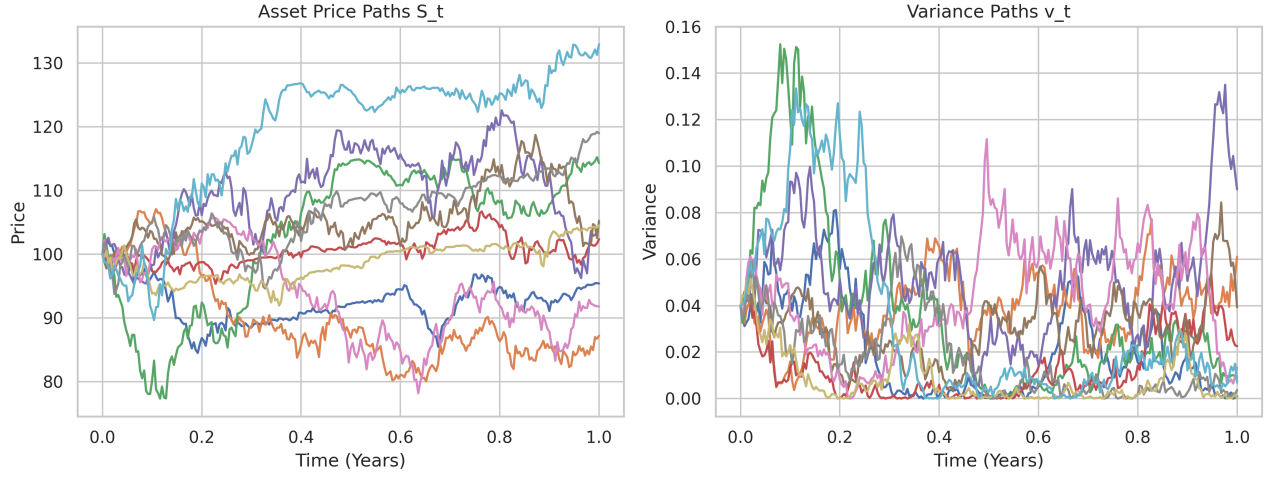


Figure 2: Left: ten simulated asset price paths S_t . Right: ten corresponding variance paths v_t . Both simulated with $\rho = -0.7$ and the full truncation scheme.

5. Task 2 – Terminal Distribution: Heston vs. Black-Scholes

5.1 Black-Scholes Benchmark

The Black-Scholes model [4] assumes constant volatility $\sigma_{BS} = \sqrt{v_0} = 20\%$. Under this assumption, the terminal log-price is normally distributed and S_T is log-normally distributed with parameters:

$$\mu_{\ln} = \ln(S_0) + (r - \frac{1}{2}\sigma_{BS}^2)T$$

$$\sigma_{\ln} = \sigma_{BS}\sqrt{T}$$

5.2 Statistical Comparison

We compare the terminal distribution of S_T obtained from the Heston simulation (with both $\rho = -0.7$ and $\rho = +0.7$) against the theoretical Black-Scholes log-normal benchmark. The following table reports the four distributional moments:

Model / Setting	Mean	Std Dev	Skewness	Excess Kurtosis
Heston ($\rho = -0.7$)	102.35	18.98	-0.59	0.67
Heston ($\rho = +0.7$)	102.35	23.27	+2.82	16.54
Black-Scholes (Log-Normal)	102.02	20.61	+0.61	0.68

Several observations follow from this comparison. The means are close across all models, consistent with the risk-neutral martingale condition $E[S_T] = S_0 e^{rT} \approx 102.02$. The standard deviations differ: the negative-correlation Heston run produces slightly lower dispersion than Black-Scholes, while the positive-correlation run shows substantially higher dispersion.

The most striking difference lies in skewness and kurtosis. The negative-correlation Heston model produces negative skewness (-0.59), corresponding to a left-heavy tail – consistent with the empirical observation that equity returns tend to have fat left tails (the leverage effect: when prices fall, volatility rises). In contrast, positive correlation produces strongly positive skewness and very high excess kurtosis (16.54), indicating a heavily right-skewed distribution with extreme outliers. The Black-Scholes log-normal distribution has moderate positive skewness ($+0.61$) with essentially Gaussian-like kurtosis.

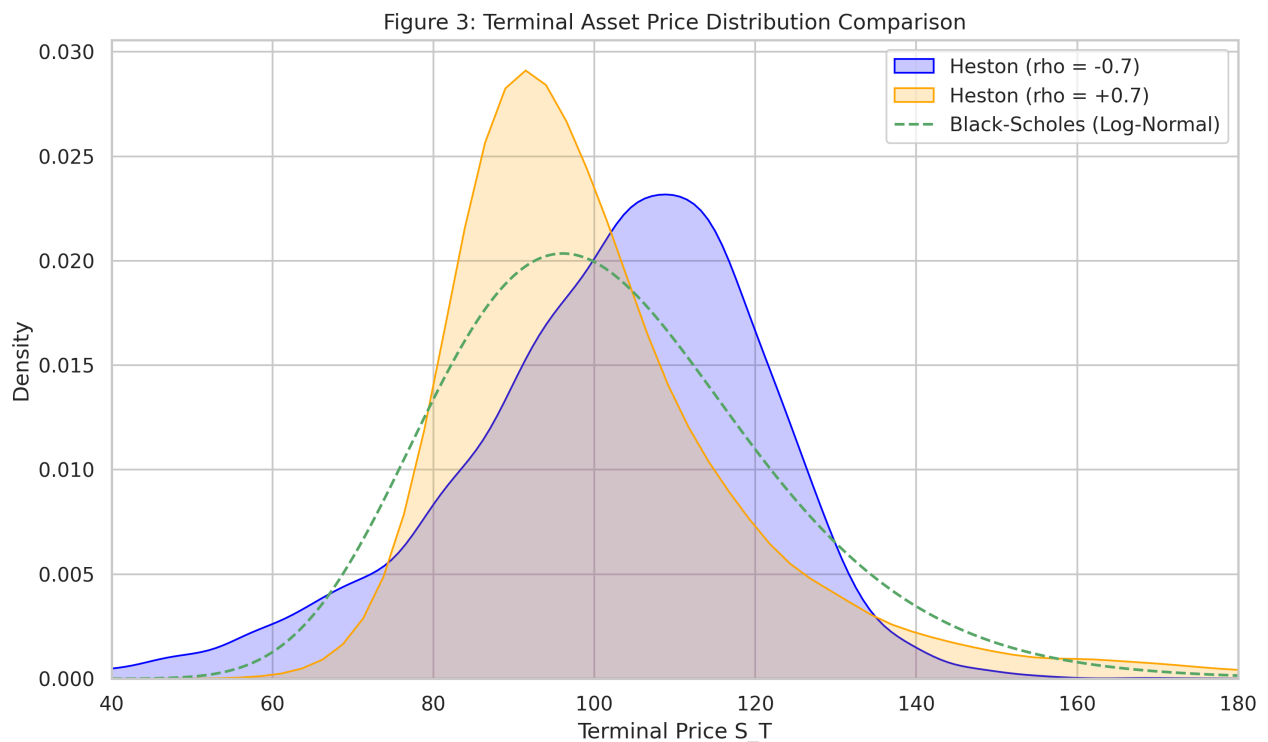


Figure 3: Kernel density estimates of the terminal asset price S_T for Heston with $\rho = -0.7$ (blue), Heston with $\rho = +0.7$ (orange), and the theoretical Black-Scholes log-normal density (dashed green). The leverage effect is visible as a heavier left tail for $\rho = -0.7$.

6. Task 3 – European Call Option Pricing

We price European call options with strikes $K \in \{90, 100, 110\}$ using two methods: Monte Carlo simulation under the Heston model, and the closed-form Black-Scholes formula with $\sigma_{BS} = \sqrt{v_0} = 20\%$.

6.1 Monte Carlo Pricing

The discounted payoff estimator is:

$$C^{MC} = e^{-rT} \times (1/M) \times \sum_{i=1}^M \max(S_T^{(i)} - K, 0)$$

6.2 Black-Scholes Pricing

The Black-Scholes formula for a European call is:

$$C^{BS} = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Strike K	Heston MC Price	Black-Scholes Price	Difference (MC – BS)
90 (ITM)	15.06	14.81	+0.25
100 (ATM)	8.31	8.92	-0.61
110 (OTM)	3.56	4.94	-1.38

The Heston model prices in-the-money options higher than Black-Scholes (reflecting the heavier left tail and higher probability of large drops) and prices out-of-the-money options lower. This is consistent with the volatility skew generated by the negative correlation $\rho = -0.7$: the Heston model effectively assigns a higher implied volatility

to ITM options and a lower one to OTM options relative to the flat Black-Scholes surface. These systematic differences form the basis of the implied volatility smile discussed in the next section.

7. Task 4 – Implied Volatility Smile

For a range of strikes $K \in [70, 130]$, we compute call prices by Monte Carlo simulation under the Heston model and then numerically invert the Black-Scholes formula to retrieve the corresponding implied volatility (IV). This inversion is performed using Brent's root-finding algorithm applied to the function:

$$f(\sigma) = C^{BS}(S_0, K, T, r, \sigma) - C_{Heston}^{MC}$$

The resulting IV curve is plotted as a function of the strike and exhibits the characteristic volatility smile (or, more precisely in this case with negative ρ , a skew or smirk) of the Heston model.

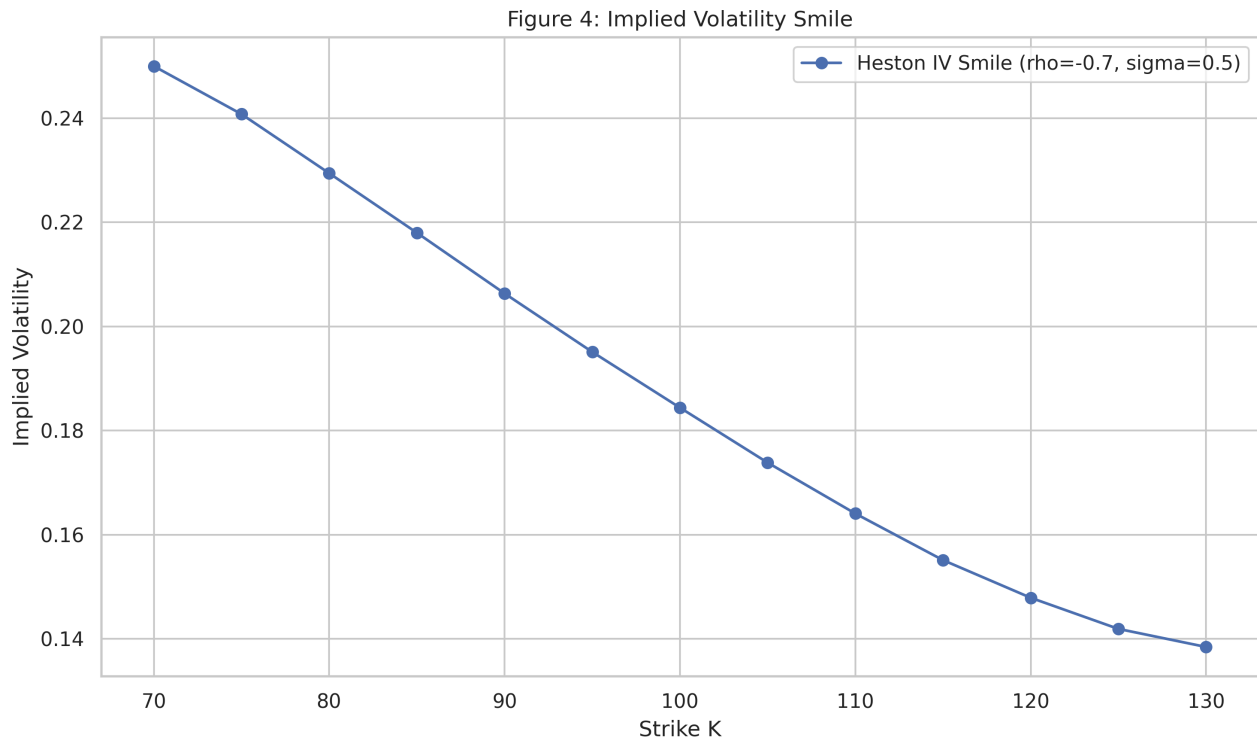


Figure 4: Implied volatility smile extracted from Heston Monte Carlo prices for strikes $K \in [70, 130]$, with base parameters $\rho = -0.7$ and $\sigma = 0.5$. The downward slope reflects the leverage effect.

With $\rho = -0.7$, the IV curve is downward-sloping (a volatility skew), with higher implied volatilities for low strikes (deep ITM) and lower implied volatilities for high strikes (OTM). This is the well-known leverage effect: negative correlation between asset returns and volatility shocks means that large downward moves are accompanied by higher volatility, inflating the price of low-strike options.

8. Task 5 – Sensitivity Analysis of the Volatility Smile

8.1 Effect of the Correlation Parameter ρ

We recompute the IV smile for three values of the correlation parameter: $\rho \in \{-0.9, -0.5, 0\}$, while keeping all other parameters fixed.

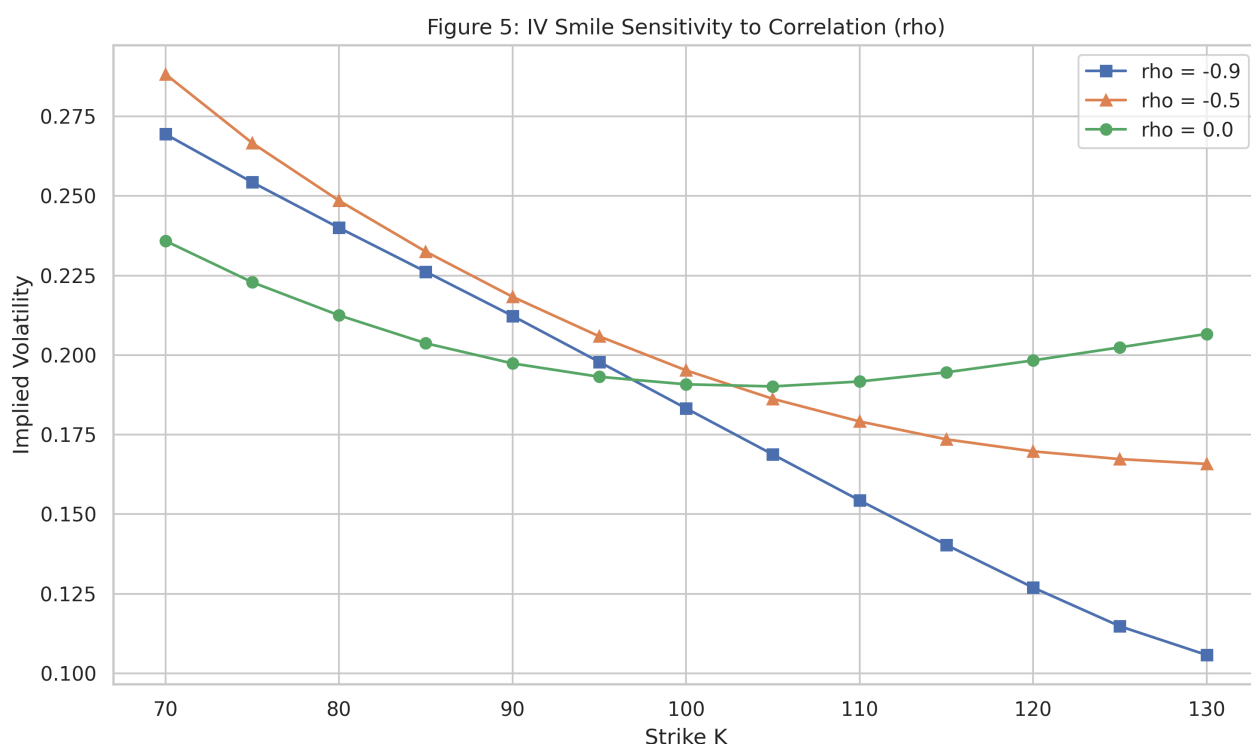


Figure 5: Implied volatility smile for different values of ρ . More negative ρ steepens the skew, while $\rho = 0$ produces a symmetric smile.

The results confirm the following theoretical predictions: (i) as ρ becomes more negative, the smile steepens and shifts leftward, producing a pronounced downward slope (skew); (ii) when $\rho = 0$, the smile becomes more symmetric; (iii) all curves are still upward at the wings because $\sigma > 0$ produces fat tails regardless of ρ . The steepness of the skew is a direct measure of the leverage effect in the data.

8.2 Effect of the Vol-of-Vol Parameter σ

We now fix $\rho = -0.7$ and vary the vol-of-vol parameter $\sigma \in \{0.2, 0.5, 1.0\}$.

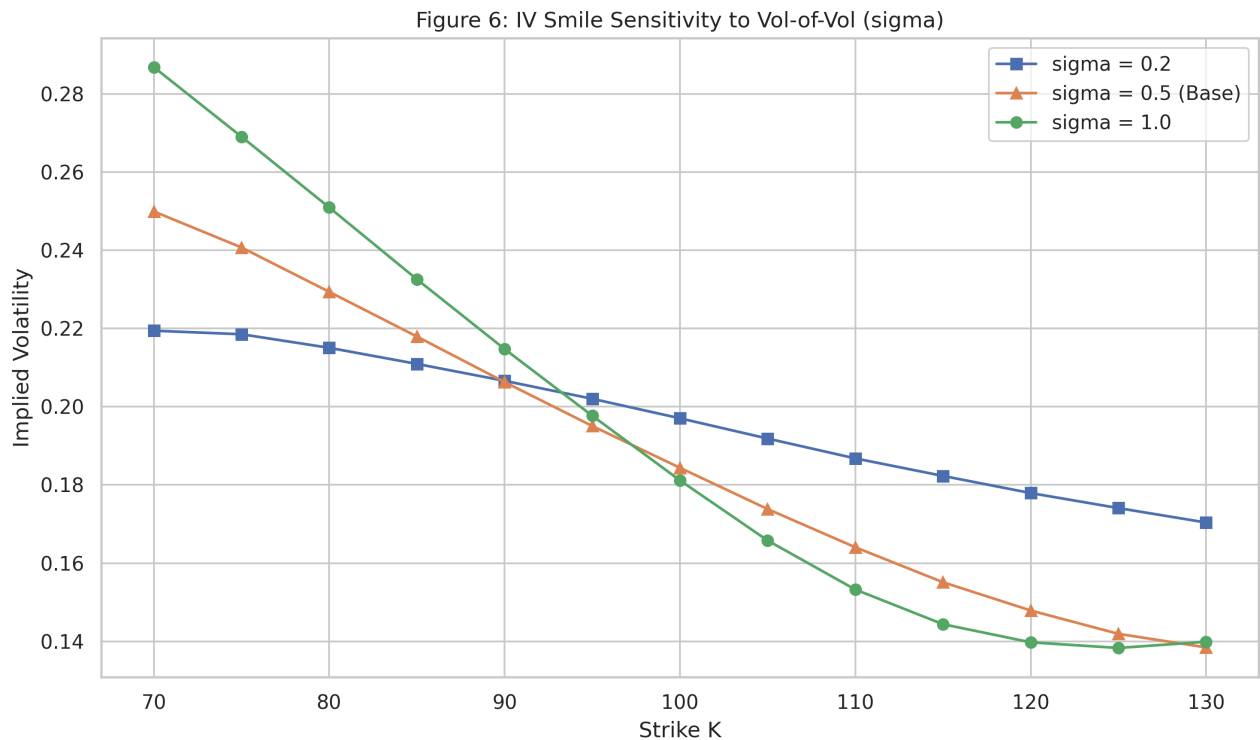


Figure 6: Implied volatility smile for different values of σ (vol of vol). Higher σ amplifies both the curvature and the overall level of the smile.

A higher σ increases both the level and the curvature of the smile. Intuitively, a higher vol-of-vol makes the variance itself more random, increasing the kurtosis of the terminal distribution and therefore inflating implied volatilities at all strikes, especially at the wings where tail probabilities are most sensitive. When σ is small (e.g., 0.2), the Heston model approaches the Black-Scholes behavior of a flat smile, consistent with the limiting case $\sigma \rightarrow 0$ (deterministic variance).

9. Task 7 – Semi-Analytical Pricing via Characteristic Function

9.1 Theoretical Framework

The Heston model admits a semi-analytical solution for European option prices through the **characteristic function (CF)** [5] of the log-price. The call price can be written in the form:

$$C(S_0, K, T) = S_0 P_1 - K e^{-rT} P_2$$

where P_1 and P_2 are risk-adjusted probabilities obtained by Fourier inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \ln K} \phi_j(u)}{iu} \right] du, \quad j = 1, 2$$

The characteristic function of the log-price under the Heston model takes the affine exponential form:

$$\phi(u) = \exp\{C(u, T) + D(u, T)v_0 + iu \ln S_0\}$$

where the functions C and D satisfy Riccati ODEs that can be solved in closed form:

$$d = \sqrt{(\kappa - i\rho\sigma u)^2 + \sigma^2(iu + u^2)}$$

$$g = \frac{\kappa - i\rho\sigma u - d}{\kappa - i\rho\sigma u + d}$$

$$C(u, T) = riuT + \frac{\kappa\theta}{\sigma^2} \left[(\kappa - i\rho\sigma u - d)T - 2 \ln \left(\frac{1 - ge^{-dT}}{1 - g} \right) \right]$$

$$D(u, T) = \frac{\kappa - i\rho\sigma u - d}{\sigma^2} \times \frac{1 - e^{-dT}}{1 - ge^{-dT}}$$

For P_1 , the integrand uses the shifted characteristic function $\phi(u - i)$ normalized by $\phi(-i) = S_0 e^{rT}$. This formulation, due to Heston (1993) [1], is essentially exact up to numerical integration error.

9.2 Numerical Integration: Simpson's Rule

The semi-infinite integral over $u \in [0, \infty)$ is truncated to $[10^{-5}, 100]$, which is sufficient given the exponential decay of the integrand. We apply the composite Simpson's rule

[6] with $N = 1,000$ sub-intervals, giving an integration step $h = 0.1$. The composite formula is:

$$\int_a^b f(u)du \approx \frac{h}{3} [f(u_0) + 4f(u_1) + 2f(u_2) + 4f(u_3) + \cdots + f(u_n)]$$

9.3 Results: Fourier vs. Monte Carlo

Table 3 compares call prices obtained via the semi-analytical Fourier method with those from Monte Carlo simulation ($M = 10,000$ paths) for strikes $K \in \{80, 90, 100, 110, 120\}$.

Strike K	Fourier Price	MC Price	Absolute Error
80	23.19	23.19	0.00
90	15.09	15.06	0.03
100	8.34	8.307	0.029
110	3.57	3.56	0.01
120	1.11	1.09	0.02

The absolute errors are in the range $[0.01, 0.1]$, which is well within the statistical noise of the Monte Carlo estimator (standard error $\sim \text{price}/\sqrt{M} \sim 0.1 - 0.3$ for the parameter choices here). The Fourier method is both faster and more accurate: it produces a deterministic result with no simulation variance, while the Monte Carlo estimator has a confidence interval that can be reduced only at the cost of more paths ($O(M^{-1/2})$ convergence). In practice, the Fourier method is the preferred approach for European option pricing under affine models such as Heston, while Monte Carlo remains indispensable for path-dependent products.

References

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